PRATEEK MISHRA

Software Engineer

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PROFESSIONAL SUMMARY

Full-Stack Software Engineer with hands-on experience building secure, scalable web platforms and immersive simulations at DRDO. Proficient in TypeScript, JavaScript, and Java across Next.js, React, Nuxt, and Node.js ecosystems. Specialized in architecting real-time collaborative systems, server-rendered consumer platforms, and classified intranet solutions with a focus on security, performance, and zero external exposure.

TECHNICAL SKILLS

Languages:	JavaScript (ES6+), TypeScript, JAVA, GO, SQL, C#
Frameworks & Libraries:	Next.js, React, Node.js, Express, Tailwind CSS
Databases:	MongoDB, PostgreSQL
Tools:	Git, Postman, Prisma, Unity, Blender, Docker, Jira, VS Code

PROFESSIONAL EXPERIENCE

Defence Research & Development Organisation (DRDO) *June 2025 – Present*
Software Engineer *Chandigarh, India*
All projects developed under confidentiality agreements for classified operational use within a government defence research organisation.

- **Architected** a full-stack intranet web portal for organizational workflow management, consolidating real-time messaging, file sharing, complaint tracking, and report generation into a single platform serving **200+ internal users**.
- **Deployed** the application on internal government servers with a focus on **security and zero external exposure**, ensuring full compliance with DRDO's data confidentiality protocols.
- **Developed** an immersive **VR Disaster Response Training Simulator** in Unity, recreating a real-world location to train personnel in emergency rescue operations, and integrated **OpenXR** standards for cross-device compatibility across VR hardware.

Technologies Used: *HTML, CSS, Node.js, Unity, OpenXR, C#*

Bilzo Labs *May 2024 – Present*
Freelance Software Engineer *Remote*

- **Delivered Wishd** for **Bilzo Labs** — a global event gifting and registry platform built on **Nuxt 3** (Vue 3 SSR), **Fastify**, and **MongoDB Atlas** in a pnpm monorepo, with SSR guest pages loading in **under 2 seconds on 4G**; deployed on **AWS**.
- **Contributed** to Bilzo Labs' consumer-facing application frontend across **[N]** release cycles, implementing UI features and integrating client-side flows against existing backend services.
- **Delivered 5+** additional full-stack applications as the sole engineer, owning requirement analysis, architecture, implementation, and deployment end-to-end.

Technologies Used: *Nuxt.js, Vue.js, React.js, Next.js, Node.js, Fastify, Express, TypeScript, MongoDB, PostgreSQL, AWS*

Om Logistics Ltd. *May 2024 – October 2024*
Quality Assurance Engineer *Noida, India*

- **Contributed** directly to the production frontend codebase of their project Rapidshyp alongside the QA mandate, implementing UI and client-side validation fixes rather than limiting involvement to defect reporting.
- **Executed** functional, regression, and **API-level** testing across release cycles for a logistics platform, authoring reproducible test cases and defect reports that reduced developer triage time.

Technologies Used: *React.js, JavaScript, HTML, CSS, REST APIs, Git, Jira, Postman*

PROJECTS

CoSketch — *Real-time Collaborative Whiteboard Application*

- **Architected** a full-stack collaborative drawing platform using **Next.js** and **WebSockets**, enabling multiple users to draw and sync shapes in real-time across shared rooms.
- **Built** a decoupled backend with separate **Express REST API** and **WebSocket** servers, handling JWT authentication, room management, and real-time shape broadcasting independently.
- **Designed** a type-safe data layer with **PostgreSQL** and **Prisma ORM**, persisting shapes as serialized JSON and replaying full room history on user join.
- **Re-engineered** the realtime server in **Go** with a lock-free, channel-based hub (goroutine-per-connection) for race-free concurrent broadcasting, deployed alongside the Node service.

Technologies Used: *Next.js, Node.js, GO, PostgreSQL, Prisma ORM, WebSockets, JWT, Zod*

EDUCATION

Chandigarh University — B.E. in Computer Science (CGPA: 8.01) *Jun 2024*
Relevant Coursework: *Data Structures & Algorithms, Operating Systems, DBMS, System Design, AI/ML, Computer Networks*

HONORS & ACHIEVEMENTS

- **GATE Qualified (CS&IT)** — Cleared the Graduate Aptitude Test in Engineering in Computer Science, a national-level competitive examination.
- **Top Scorer, AMCAT Assessment** — Achieved the highest score in class, demonstrating strong aptitude in technical and analytical evaluation.